

NIEN-CHEN (JANE) WU

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EDUCATION

University of California, Irvine

Master of Science in Software Engineering – GPA:3.95/4.0

Sept. 2023 – Dec. 2024

Irvine, CA

National Chengchi University

Bachelor of International Business

Sept. 2017 – Jun. 2021

Taipei, Taiwan

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, Node.js, HTML/CSS, SQL

Frameworks & Databases: Spring, Django, React, Express, Next.js, RESTful API; MySQL, PostgreSQL, MongoDB, NoSQL

Tools: AWS(S3, EC2, Lambda, DynamoDB, IAM, Cognito), JUnit, Jest, Git, Docker, Jenkins, Maven, Postman, Jira

Software Engineering: Agile/Scrum, Software Development Lifecycle, Unit Testing, CI/CD, Version Control, Code Review

Soft skills: Problem-Solving, Teamwork, Communication, Critical Thinking, Quick Learning, Time Management

WORK EXPERIENCE

Prenostik (Startup)

Software Engineer Intern (Full-stack)

Sept. 2024 – Dec. 2024

Irvine, CA, USA

- Developed a SaaS student dashboard using **React**, **Next.js**, and **TypeScript** to help over 300 students analyze learning challenges and improve learning motivation.
- Implemented server-side rendering, static site generation and **Redis** caching to improve SEO and reduce load time by 1.5s.
- Optimized **PostgreSQL** queries by indexing and caching, reduced API response times by 40% and improved data retrieval.
- Collaborated with cross-functional teams to deploy **Java Spring Boot microservices** on **AWS** using **Docker**, improving system scalability and enabling real-time student analytics.

UC Irvine Donald Bren School of Information and Computer Sciences

Software Engineer Intern (Backend)

Jun. 2024 – Sept. 2024

Remote, USA

- Built an image deduplication module in **Java** using MD5 hashing to detect duplicates with 0.5s processing time per image.
- Stored image metadata in **PostgreSQL** and optimized indexing strategies to reduce query execution time by 40%.
- Designed a serverless image processing pipeline with **AWS Lambda** and **S3 Event Notifications** to automate S3 uploads, format conversion, metadata extraction, and resolution checks, reducing image processing time from 10s to 4s per image.
- Optimized API response times by managing image metadata in **DynamoDB** with partition keys and TTL-based auto-deletion, collaborating with database engineers to refine data retrieval, reduce storage costs, and prevent duplicates.
- Implemented a resilient event-driven architecture with **AWS Dead Letter Queues** and **CloudWatch** for error logging, and automatic retries, improving pipeline reliability through collaboration with cloud engineers.

ASUS

Data Analyst

Jul. 2021 – Mar. 2022

Taipei, Taiwan

- Automated large-scale data processing workflows using **Python (Pandas, NumPy)**, reducing analysis time by 25%.
- Collaborated with software engineers to optimize the website navigation bar, resulting in a 30% increase in user CTR.

Microsoft

Product Assistant Intern

Jul. 2020 – Jun. 2021

Taipei, Taiwan

- Analyzed product usage using Excel and **Python** to identify low-engagement accounts, supporting re-engagement strategy.
- Organized and led a campus tour campaign across 15 universities, driving 1,000+ new LinkedIn group memberships.

PROJECTS

Microservice Application | Backend | Java, Spring Boot, REST

- Built and deployed a secure **RESTful** microservice using **Java** and **Spring Boot** with **JWT**-based user authentication.
- Integrated inter-service communication with **Eureka** for service discovery and **Spring API Gateway** for load balancing.
- Containerized microservices with **Docker** and deployed on **AWS EC2**, enhancing scalability and reliability.

Insecure Bank Web Application | Backend | Python, Flask

- Developed and tested backend user authentication and transaction logic using **Python**, **Flask**, and **SQLAlchemy**.
- Identified and resolved security vulnerabilities based on formal assessment reports, enhancing app resilience.

Mailbox Web page | Full-stack | JavaScript, React, Node.js

- Designed and implemented responsive front-end mailbox interfaces using **HTML**, **CSS**, **React**, and **JavaScript**.
- Built backend APIs with **Node.js** to support email sending, retrieval, and contact management features.